

8 (short)

(a) A memory chip has 30 address wires and 8 data wires. Calculate its capacity in bits. State also the capacity in megabytes. [3]

(b) (i) Consider the following commands in PIC assembly language :

```
        movlw 0x02
        movwf 0x20
lb      decfsz 0x20
        goto lb
```

Describe how the contents of the memory location 0x20 change as the commands above are executed. [4]

(ii) Describe how the contents of the memory location 0x20 change as the commands below are executed instead :

```
        movlw 0x02
lb      movwf 0x20
        decfsz 0x20
        goto lb
```

[3]